

CSE 370 – Database Systems

Assignment 03

Full Name (in Block Letter): Imtiaz Hossain

ID: 23101137

Question 1 [CO6] : 5 Points

A PARTS file with Part# as the key field includes records with the following Part# values: 63, 35, 58, 23, 59, 70, 30, 25, 92, 86. Suppose that the search field values are inserted in the given order in a B+ tree of order $n = 4$; show how the tree will expand and what the final tree will look like.

After building the tree in this sequence, delete these values: 35, 86, 63, 92, 30 and show how the tree will reduce and what the final tree will look like.

[63]

[35 | 63]

[35 | 58 | 63]

[58]
[23 | 35] → [58 | 63]

[58]
[23 | 35] → [58 | 59 | 63]

[58 | 63]
[23 | 35] → [58 | 59] → [63 | 70]

[58 | 63]
[23 | 30 | 35] → [58 | 59] → [63 | 70]

[30 | 58 | 63]
[23 | 25] → [30 | 35] → [58 | 59] → [63 | 70]

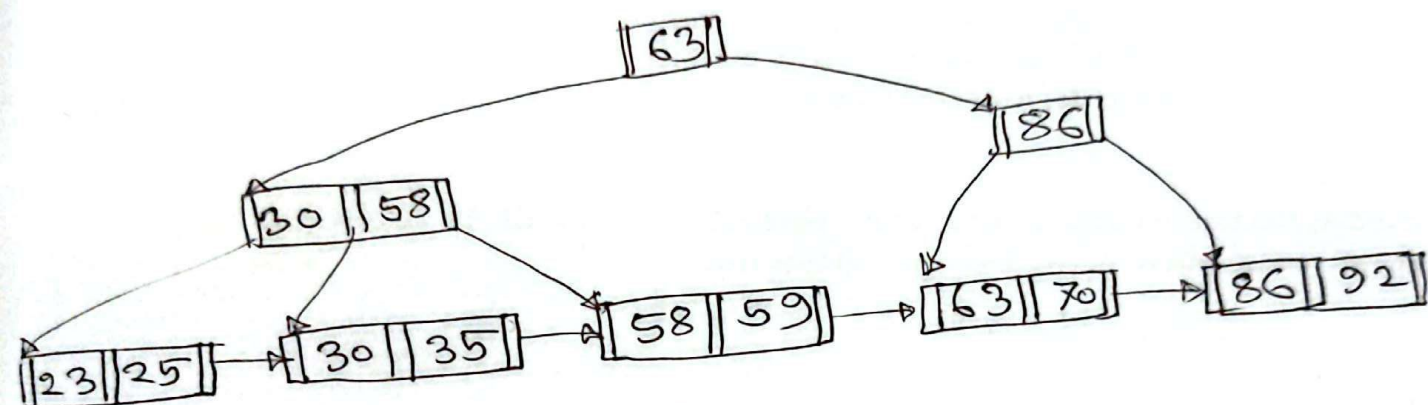
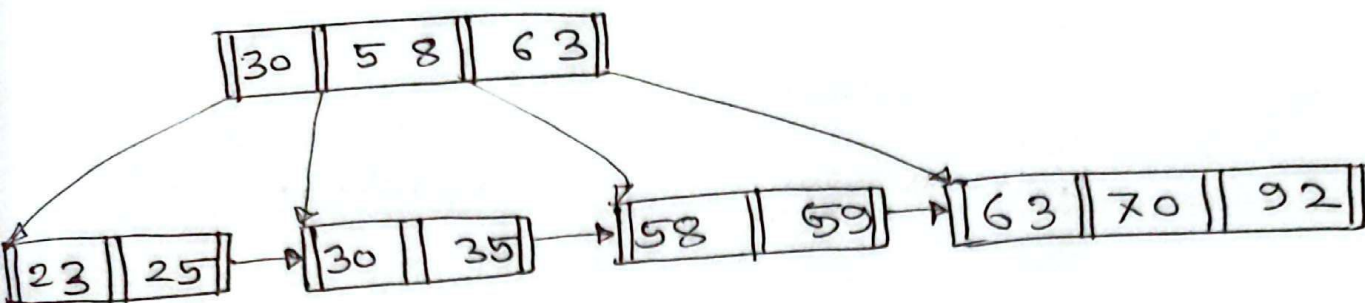
$$n=4$$

$$\text{minimum} = \lceil n/2 \rceil - 1$$

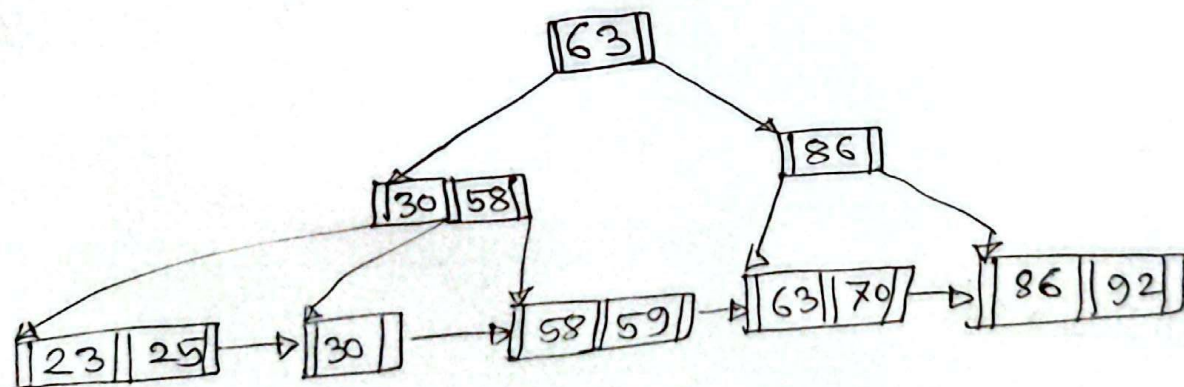
$$= 2 - 1 = 1$$

$$\text{maximum} = n - 1$$

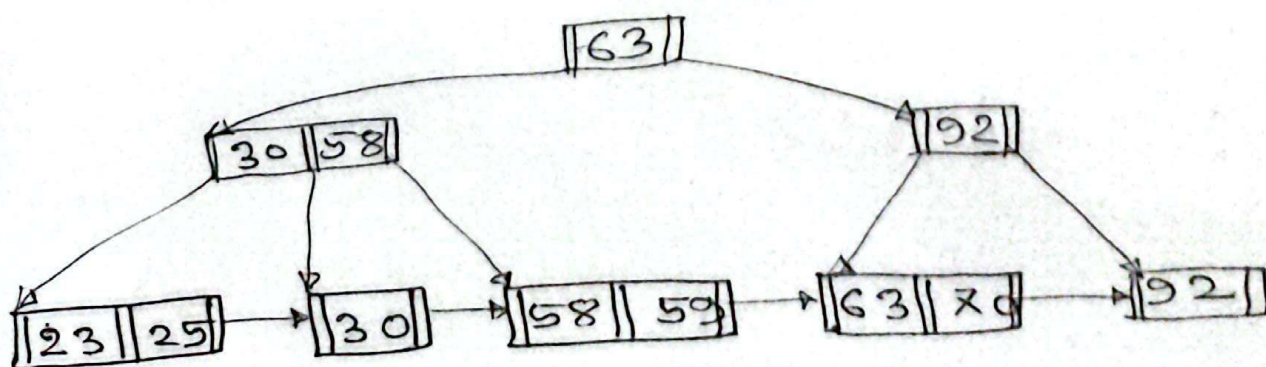
$$= 4 - 1 = 3$$



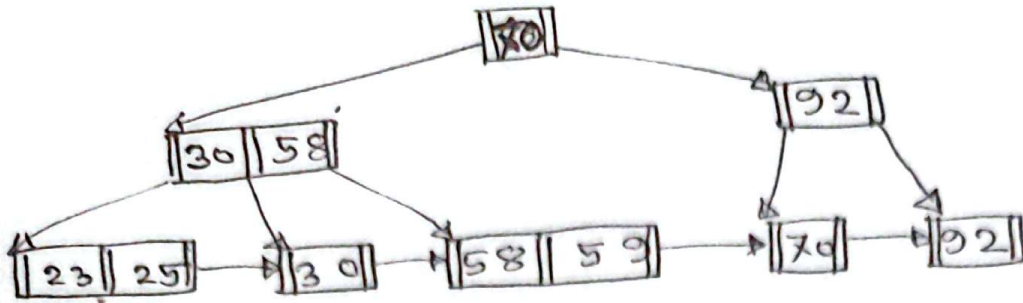
Deleting 35



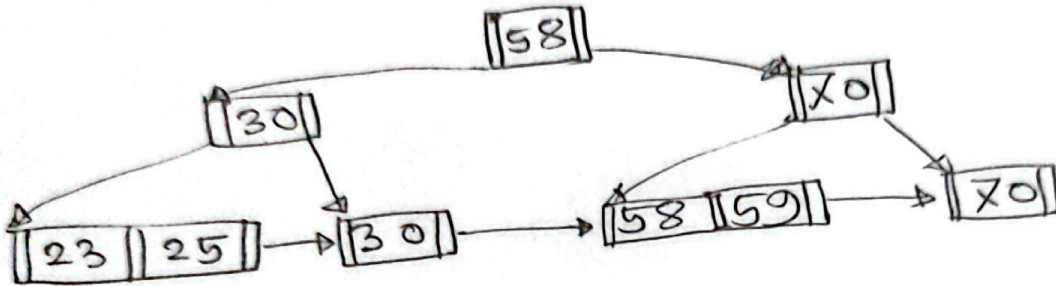
Deleting 86



Deleting 63



Deleting 92



Deleting 30

